

Exponential growth and decay - 2.8

To say that some function $y(t)$ grows exponentially means y is of the form $y(t) = y_0 e^{kt}$.

Seen this before from compound interest

$$A(t) = A e^{rt}$$

This covers all bases: $e^{kt} = (e^k)^t = a^t$

where $a = e^k \iff k = \ln(a)$

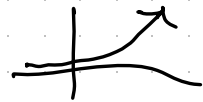
Prefer to think in terms of derivatives:

$$y'(t) = \frac{d}{dt} y_0 e^{kt} = y_0 e^{kt} \cdot k = ky(t)$$

To say that $y(t)$ grows exponentially is
to say that $\boxed{y' = ky}$

And we call k the growth constant.

Exponential growth when $k > 0$,



Exponential decay when $k < 0$



Example: Suppose we have a population of 200 bacteria with a growth constant of .02. Then at time t , there are $200e^{.02t}$.

How long does it take for the population to reach 100,000 bacteria?

$$\rightarrow 200e^{.02t} = 100000 \Leftrightarrow e^{.02t} = 500$$

$$.02t = \ln(500) \Leftrightarrow t = \frac{\ln(500)}{.02} = 50 \ln(500)$$

$$\approx 310.7 \text{ minutes}$$

How long does it take for something growing exponentially to double?

$$y_0 e^{kt} = 2y_0$$

— can divide by y_0 /
the time needed doesn't
depend on the initial
size y_0

$$e^{kt} = 2$$

$$kt = \ln(2)$$

$$t = \frac{\ln(2)}{k}$$

"doubling time"

If k is negative, then we call this
a half-life instead — time it takes to halve

$$\begin{aligned} y_0 e^{kt} &= \frac{1}{2} y_0 \iff e^{kt} = \frac{1}{2} \iff kt = \ln\left(\frac{1}{2}\right) \iff t = \frac{\ln\left(\frac{1}{2}\right)}{k} \\ &= \frac{\ln(1) - \ln(2)}{k} = \frac{-\ln(2)}{k} = \frac{\ln(2)}{|k|} \end{aligned}$$

Example: Carbon-14 has a half-life of 5730 years. If an artifact has 10g of Carbon-14 and we know it started with 100g, how old is it?

$$C(t) = 100e^{-kt}$$

$$\frac{\ln(2)}{k} = 5730$$

$$\Leftrightarrow k = \frac{\ln(2)}{5730}$$

Looking for t when

$$10 = 100e^{-\frac{\ln(2)}{5730}t}$$

$$\Leftrightarrow 1 = 10e^{-\frac{\ln(2)}{5730}t} \Leftrightarrow e^{\frac{\ln(2)}{5730}t} = 10$$

$$\frac{\ln(2)}{5730}t = \ln(10) \Leftrightarrow t = \frac{\ln(10) \cdot 5730}{\ln(2)} \approx 19034.6 \text{ years.}$$